RIVER VALLEY HIGH SCHOOL
H2 CHEMISTRY

Suggested Solutions for 2011 A Level H2 Chemistry Paper 1

Question Number	Key	Question Number	Key
1	A	21	C
2	D	22	B
3	C	23	B
4	D	24	B
5	C	25	C
6	C	26	A
7	D	27	A
8	C	28	D
9	A	29	B
10	D	30	B
11	C	31	D
12	B	32	B
13	B	33	A
14	B	34	B
15	C	35	C
16	C	36	B
17	C	37	D
18	A	38	D
19	C	39	A
20	B	40	B

Suggested Solutions for 2011 A Level H2 Chemistry Paper 2

1 (a) (i) $HOC_6H_4CO_2H + (CH_3CO)_2O \rightarrow CH_3CO_2C_6H_4CO_2H + CH_3CO_2H$

(ii) $\eta_{aspirin} = \frac{10}{180} = 0.0556$ mol

$\eta_{2\text{-hydroxybenzenecarboxylic acid}}$ required = $\eta_{\text{ethanoic anhydride}}$ required

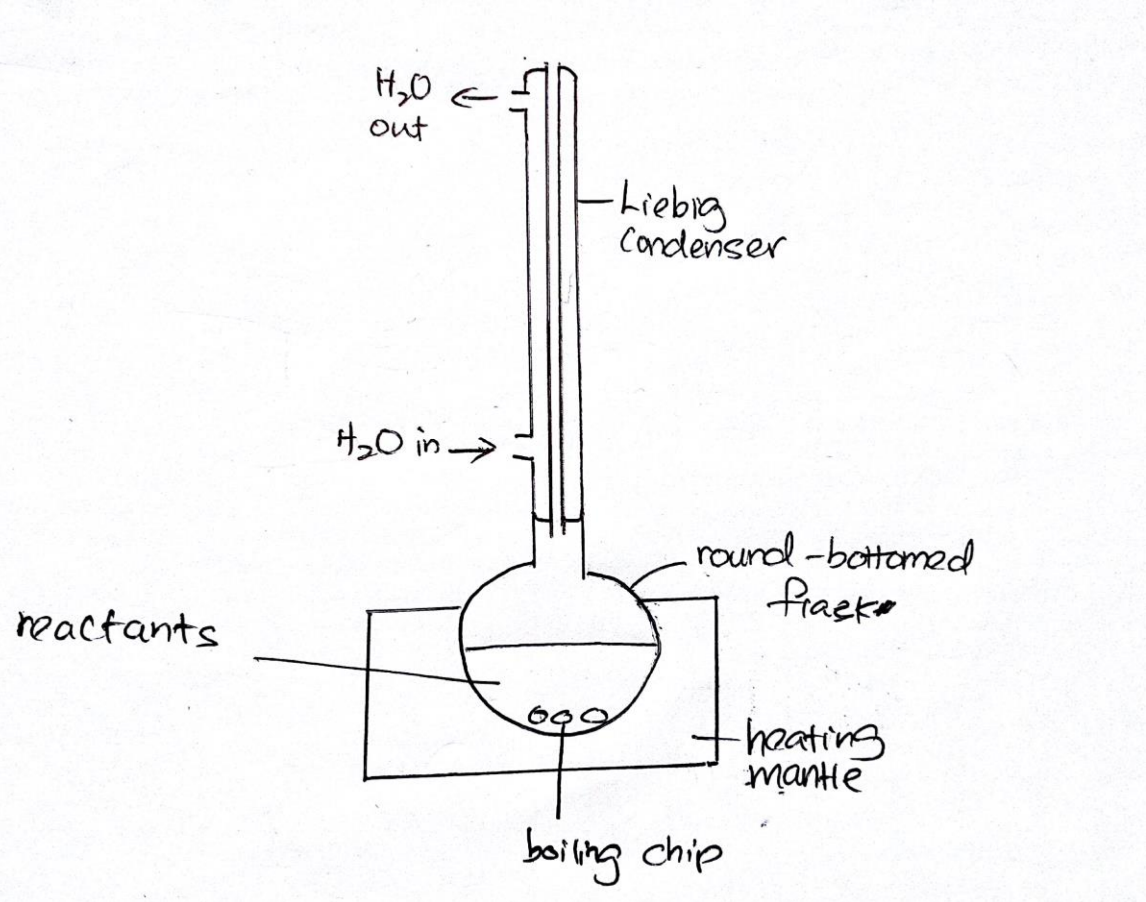
$= 0.0556 \times \frac{100}{75} = 0.0741$ mol

mass of 2-hydroxybenzenecarboxylic acid required = 0.0741×138

= **<u>10.2 g</u>**

mass of ethanoic anhydride required = 0.0741×102 = **<u>7.56 g</u>** [3]

(b)



(- Do NOT seal the top of the Liebig condenser with a stopper / thermometer. Build up of pressure from the vapour will be dangerous!

- Use a round bottom flask for even heating; do NOT use a conical flask / beaker which will result in uneven heating.

- Use a heating mantle; do NOT use a naked flame from the bunsen burner which will be dangerous since organic compounds, e.g. ethanoic anhydride, are flammable.) [8]

1. Weigh a **dry and empty** 100 cm^3 beaker.
2. Weigh **accurately about 10.2 g** of 2-hydroxybenzenecarboxylic acid into the beaker.
3. **Transfer** the solid into a **250 cm^3 / 100 cm^3 round bottom flask**.
(No need to reweigh the 100 cm^3 beaker, since the exact mass of solid added is not important in this experiment.)

4. Place the round bottom flask in an **ice bath**.

   (The question mentioned that the initial reaction may be violent, hinting that safety is of concern. Hence, lowering the temperature to slow down the initial rate of reaction is required.)

5. Using a **10 cm^3 measuring cylinder**, measure (7.56 g / 1.08 g cm^{-3} =) **7 cm^3 of ethanonic anhydride** add it into the round bottom flask containing the 2-hydroxybenzenecarboxylic acid **slowly**, **using a dropper**.

   (Ethanoic anhydride is a corrosive liquid and weighing on a balance needs to be avoided.

   A dropper is used for transferring to prevent direct contact with the corrosive liquid.)

6. Add **dropwise**, 8 to 10 drops of 85% phosphoric acid into the same round bottom flask, **using a dropper**.

7. Remove from the ice bath and **set up the reflux set-up** shown in the above diagram and **heat** the reaction mixture for around **15 minutes**.

8. Switch off the heating mantle.

9. Using a **10 cm^3 measuring cylinder**, measure **2-3 cm^3 of water** and add it into the round bottom flask **slowly**, **using a dropper**.

   (The question mentioned that addition of water may cause the mixture to boil, hinting that safety is of concern. Hence, highlight that water must be added slowly using a dropper.)

10. Allow the reaction mixture to **cool down to room temperature**.

11. Using a **50 cm^3 measuring cylinder**, measure **50 cm^3 of cold water** and pour it into a **100 cm^3 conical flask / beaker**.

12. Pour the reaction mixture in the round bottom flask into the beaker containing the cold water.

13. **Filter** the mixture and **collect the crude product** as the **residue**.

14. Transfer the crude product into a clean **100 cm^3 / 250 cm^3 conical flask**.

15. Add a **small volume of water** and **heat** the mixture until **all the crude product just dissolves**. Add **small volumes of hot water when necessary** to dissolve all the crude product.

    (Excessive amounts of water should not be added so as to ensure that a hot saturated solution is obtained. A saturated solution is required for crystallisation to occur upon cooling later.)

16. Using a **pre-warmed dry filter funnel** and **fluted filter paper**, **filter the hot solution** into a **pre-warmed clean and dry 100 cm^3 / 250 cm^3 conical flask**.

    (This step is carried out to remove insoluble impurities in the hot solution.

    The filtration apparatus should be pre-warmed to ensure that crystals don't form during the filtration process, which will otherwise clog up the filter funnel.

    The apparatus also need to be dry to ensure that the hot solution remains saturated. If the solution gets diluted, crystallisation may not occur upon cooling.)

17. Allow the solution to **cool down to room temperature** slowly for crystals to form.

18. When the solution is cooled to room temperature, submerge the conical flask into an **ice bath** to allow all the pure product to precipitate.

19. **Filter** the solution to obtain the pure aspirin as the **residue**.

    (The soluble impurities will remain in the filtrate.)

20. **Dry** the crystals under an **infra-red lamp / pressing it between pieces of filter paper**.

21. To check the purity of the aspirin, **determine the melting point** of the crystals collected. A pure sample of aspirin should melt with a **narrow temperature range** close to **135 °C** (melting point of aspirin).

(Recall that we learnt in Year 6, Practical 9 (Hydrolysis of methylbenzoate Annex 1)

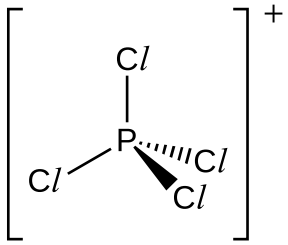
(c) Ethanoic anhydride is corrosive so gloves should be worn to avoid contact with skin. *OR*

Ethanoic anhydride is volatile and toxic so experiment should be performed in a fume cupboard. *OR*

Ethanoic anhydride used is flammable so avoid having naked flames around the reaction set-up. [1]

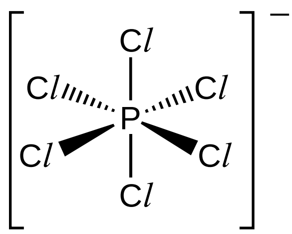
2 (a) $$K_C = \frac{[PCl_3][Cl_2]}{[PCl_5]}$$ Units: mol dm^{-3} [2]

(b) $[PCl_4]^+$:



Tetrahedral

$[PCl_6]^-$:



Octahedral [3]

3 (a)

	$PCl_5(g)$	$\rightleftharpoons$	$PCl_3(g)$	+	$Cl_2(g)$
$\eta_{initial}$ / mol	1.00		0		0
$\eta_{equilibrium}$ / mol	1.00 – x		x		x

η_{H^+} formed from PCl_3 = x(3 + 2) = 5x mol

η_{H^+} formed from PCl_5 = (1–x)(5 + 2) = (7 – 7x) mol

Total η_{H^+} formed from mixture of PCl_5 & PCl_3 = 5x + (7 – 7x) = **<u>(7 – 2x) mol</u>** [4]

(b) (i) η_{H^+} in 25.0 cm^3 of standard solution = (37.4/1000) × 4.00 = 0.150 mol

η_{H^+} in 1 dm^3 of standard solution = 0.150 × (1000/25) = **<u>6.00 mol</u>**

(ii) 7 – 2x = 6.00

∴ x = 0.500 mol

Equilibrium η_{Cl_2} = Equilibrium η_{PCl_3} = x = **<u>0.500 mol</u>**

Equilibrium η_{PCl_5} = 1 – x = **<u>0.500 mol</u>** [4]

(c) $$K_C = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{\left(\frac{0.500}{5}\right)^2}{\frac{0.500}{5}} = \mathbf{0.100\ mol\ dm^{-3}}$$ [2]

4 (a) (i) They conduct electricity in the solid state. / They are lustrous/ductile/ sonorous/malleable.

(ii) **Boiling point** **increases** down the group from chlorine to iodine.

Down the group, the **number of electrons** in the molecule **increases**, hence **more energy** is required to **overcome** the **stronger instantaneous dipole-induced dipole interactions** **between** the **molecules**, resulting in an increase in boiling point. [4]

(b) (i) Giant covalent structure

(ii) From carbon to germanium, the **size of atoms increases** and this results in a **decrease in effectiveness of overlap of the orbitals**. The **strength of the covalent bond decreases** from carbon to germanium. Amount of **energy** needed to **break the covalent bonds** **decreases** from C−C to Ge−Ge bonds, hence **melting point** **decreases** in the same order. [3]

(c) (i) $SiCl_4 + 2H_2O \rightarrow SiO_2 + 4HCl$

OR

$SiCl_4 + 4H_2O \rightarrow SiO_2.2H_2O + 4HCl$

(Wrong: $Si(OH)_4$)

(ii) C−C*l*: +340 kJ mol^{-1}

Si−C*l*: +359 kJ mol^{-1}

(iii) Carbon atom does not have **energetically accessible** **vacant orbitals** for **dative bonding with water** (as the 3s, 3p and 3d orbitals are too high in energy to be available for dative bonding). Hence CCl_4 is inert to water. [3]

(d) (i)

	C*l*	H	N	Pb
Mass ratio	46.7	1.76	6.14	45.4
Molar mass	35.5	1.0	14.0	207
Mole ratio	46.7/35.5 = 1.32	1.76/1.0 = 1.76	6.14/14.0 = 0.439	45.4/207 = 0.219
Simplest ratio	1.32/0.219 = 6.03	1.70/0.219 = 8.04	0.439/0.219 = 2.00	0.219/0.219 = 1.00
	6	8	2	1

Empirical formula of **Y** is $Cl_6H_8N_2Pb$

(d) (ii) Let the relative formula of **Y** be $(Cl_6H_8N_2Pb)_n$.

$456 = n[(14.0 \times 2) + (1.0 \times 8) + (35.5 \times 6) + 207]$

n = 1

Formula of **Y** is $Cl_6H_8N_2Pb$

Since **Y** contains 1 anion and 1 cation in the ratio of 1:2,

Anion: $[PbCl_6]^{2-}$

Cation: NH_4^+

(Hint: Since the NH_4Cl solution contains NH_4^+ ions, it can be deduced that **Y** contains NH_4^+ as the cation. Since there are 2 nitrogens in the formula, it implies that there are 2 NH_4^+ cations. The formula of the anion can then be deduced accordingly.)

(iii) Oxidation state of Pb in **Y**: +4

(iv) Octahedral (coordination number of 6) [5]

(e)

$Pb^{4+} + 2e^- \rightleftharpoons Pb^{2+}$ $E^\ominus = +1.69$ V

$VO^{2+} + 2H^+ + e^- \rightleftharpoons V^{3+} + H_2O$ $E^\ominus = +0.34$ V

$VO_2^+ + 2H^+ + e^- \rightleftharpoons VO^{2+}$ H_2O $E^\ominus = +1.00$ V

$VO_3^- + 4H^+ + e^- \rightleftharpoons VO^{2+}$ $2H_2O$ $E^\ominus = +1.00$ V

Oxidation of V^{3+} to VO^{2+} by Pb^{4+}:

$Pb^{4+} + 2V^{3+} + 2H_2O \rightarrow 2VO^{2+} + 4H^+ + Pb^{2+}$

$E^\ominus_{cell} = +1.69 - (+0.34) = +1.35 \text{ V} > 0 \text{ V}$ (energetically feasible)

Pb^{4+} can oxidise V^{3+} to VO^{2+}.

Further oxidation of VO^{2+} to VO_2^+ by Pb^{4+}:

$Pb^{4+} + 2VO^{2+} + 2H_2O \rightarrow 2VO_2^+ + 4H^+ + Pb^{2+}$

$E^\ominus_{cell} = +1.69 - (+1.00) = +0.69 \text{ V} > 0 \text{ V}$ (energetically feasible)

Pb^{4+} can further oxidise VO^{2+} to VO_2^+.

OR

Further oxidation of VO^{2+} to VO_3^- by Pb^{4+}:

$Pb^{4+} + 2VO^{2+} + 4H_2O \rightarrow 2VO_3^- + 8H^+ + Pb^{2+}$

$E^\ominus_{cell} = +1.69 - (+1.00) = +0.69 \text{ V} > 0 \text{ V}$ (energetically feasible)

Pb^{4+} can further oxidise VO^{2+} to VO_3^-.

(Using $E^\ominus_{cell}$ values, it may seem like Pb^{4+} will oxidise Cl^- to Cl_2. However, this cannot be true since a solution of $PbCl_4$ exists in this question, which implies that a solution of Pb^{4+} and Cl^- ions is stable.) [2]

5 (a) (i) Since **K** contains only 4 carbon atoms and decolourises aqueous bromine, **K** undergoes **electrophilic addition** and has **1 carbon-carbon double bond** (C=C bond) / alkene functional group.

(- **K** is not a phenol as it has only 4 carbon atoms, hence it could not have undergone electrophilic substitution with aqueous bromine.

- There is probably only 1 C=C bond based on the formula of compound **K**.)

(ii) Since **K** reacts with sodium metal to form hydrogen gas.

Since **K** has only 4 carbon atoms, it is not a phenol.

Hence, **K** is either an alcohol or a carboxylic acid.

$$ROH + Na \rightarrow RO^{-}Na^{+} + \frac{1}{2}H_2$$

$$RCOOH + Na \rightarrow RCOO^{-}Na^{+} + \frac{1}{2}H_2$$

1 mol of alcohol or acid functional group reacts with sodium in an **acid-metal reaction** to form **½ mol of hydrogen gas**.

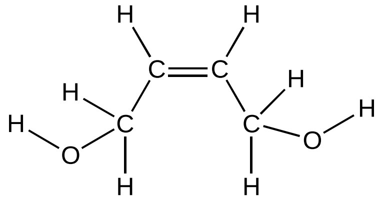
Amount of H_2 formed = 2.4 ÷ 24 = 0.100 mol

Since 0.1 mol **K** forms 0.1 mol of hydrogen gas, there should be **2 alcohol or carboxylic acid groups** in 1 molecule of **K**.

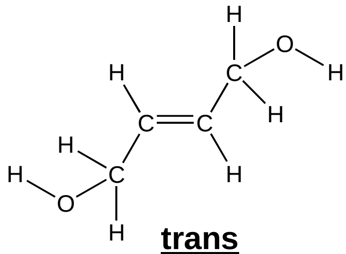
If there are 2 carboxylic groups in **K**, there would be 4 oxygen atoms. Hence, the presence of **2 oxygen atoms** in the molecular formula suggests that **K** contains **2 alcohol groups** in a molecule. [5]

(b) Cis-trans isomerism:

Arises due to **restricted rotation about C=C bond**



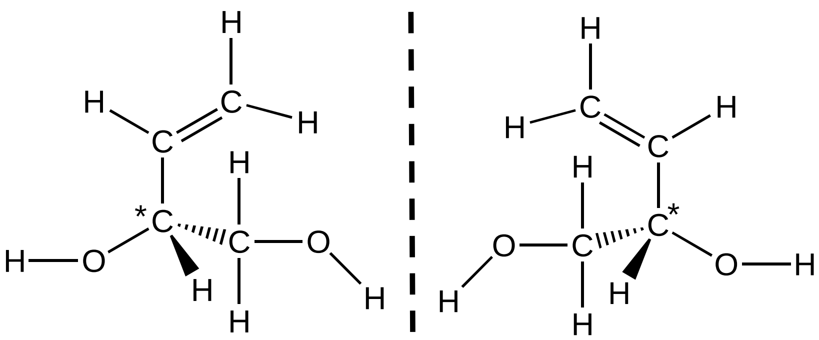
cis



trans

Enantiomerism:

Enantiomers can **rotate plane-polarised light**



* **denotes chiral carbon**

(A displayed formula is required - show the bond angles for trigonal planar and bent centres.) [7]

6 (a) (i) Silver chloride ($AgCl$)

(ii) Alkyl chloride

(Reaction 1 involves the hydrolysis of alkyl chloride by heating with H_2O, producing Cl^- which then precipitates out as $AgCl$.)

(iii) Phenol and alcohol

(Wrong: carboxylic acid – since there is only 1 oxygen atom in **W**)

(iv) Phenol

Phenol undergoes **acid-base reaction** with NaOH(aq) at room temperature to form the sodium phenoxide **salt**, which is ionic and **soluble in water** to form a colourless solution.

Alcohol, being a **weak acid**, **does not react** with NaOH(aq). [5]

(b) $C_7H_6O_3$ (benzene ring with a –COOH and a –OH group attached to it) [1]

(c) (i) $C_7H_4OClBr_3$

(3 hydrogen atoms in phenol substituted by 3 bromine atoms)

(ii) Electrophilic substitution

(iii) The compound is formed from the electrophilic substitution of **aromatic phenol** where the **−OH group activates the benzene ring** towards electrophilic substitution. **Aliphatic compounds do not undergo electrophilic substitution** reaction.

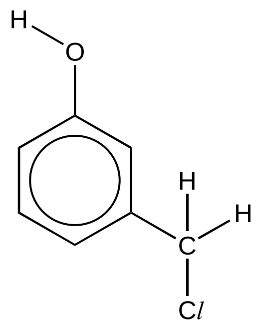
(iv) Phenol

(v) Reaction 3

Side chain oxidation of $-CH_2Cl$ gives the hydroxybenzoic acid with M_r of 138.

(Wrong: Reaction 5 – reaction 5 on its own, without the molecular formula given, cannot confirm that a benzene ring is present, since a carboxylic acid will also give the same result) [6]

(d) (i)



(A displayed formula is required - show the bond angles for trigonal planar and bent centres.) [4]

(ii) Reaction 2 produces the **tri-bromo substitution** product with M_r of 379.2. **Electrophilic substitution** results in the **Br atoms** being substituted at **positions 2, 4 and 6** relative to the **−OH group** since it is **2,4-directing**. Hence the **$-CH_2Cl$ group would be at position 3** relative to the −OH group.

Reaction 1 suggests that **W** can undergo **nucleophilic substitution** to produce Cl^- that forms a white precipitate with Ag^+. Hence, **W** is an **alkyl halide** and **not an aryl halide**.

(Note: Must explain the position of the –OH group and the –C*l* group.)

Suggested Solutions for 2011 A Level H2 Chemistry Paper 3

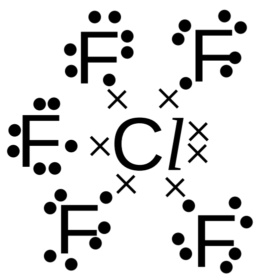
1 **(a)** Let the oxidation number of iodine in ICl_4^- be x.

$x + 4(-1) = -1$

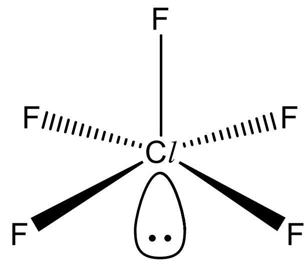
$x = +3$

(C*l* is more electronegative than I. Hence, C*l* is the one having an oxidation state of –1, while I is assigned a positive oxidation state.) [1]

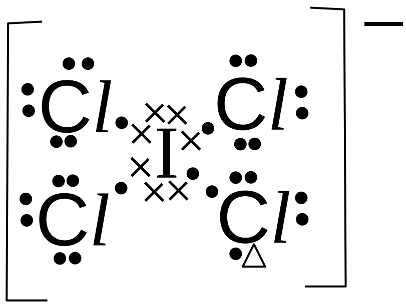
(b)

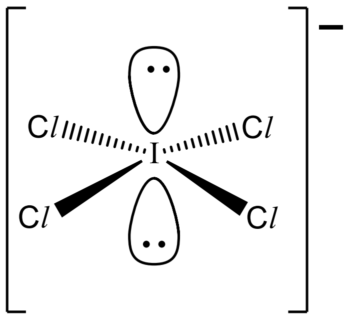


5 bond pairs, 1 lone pair



square pyramidal





square planar

4 bond pairs, 2 lone pairs

(Note: In negatively charged ICl_4^-, one of the C*l* will have an extra electron since it is more electronegative than I. The resultant Cl^- will then bond with the I atom through dative bonding.) [4]

(c) **(i)** $HClO + H^+ + 2I^- \rightarrow Cl^- + I_2 + H_2O$

(- Construct the respective half equations (HC*l*O reduced to Cl^-, I^- oxidised to I_2), then combine to give the overall equation.

- Construct the equations under acidic conditions, since it is stated in the question that the KI used is *acidified*.)

(ii) $n_{Na_2S_2O_3}$ in 6.0 cm^3 $= \dfrac{6.0}{1000} \times 0.0050$

$= 3.00 \times 10^{-5}$ mol

n_{I_2} produced $= 3.00 \times 10^{-5} \times \dfrac{1}{2}$

$= 1.50 \times 10^{-5}$ mol

n_{HClO} produced = $n_{N\text{-chlorosuccinimide}}$ in tablet = 1.50×10^{-5} mol

mass of *N*-chlorosuccinimide

$= 1.50 \times 10^{-5} \times (4 \times 12.0 + 2 \times 16.0 + 14.0 + 35.5 + 4 \times 1.0)$

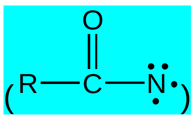
$= 2.00 \times 10^{-3}$ g [4]

(d) **(i)** $R–CONH_2 + Br_2 \rightarrow R–CONHBr + HBr$

(ii) +1

(Br is covalently bonded to N. Since N is more electronegative, Br has a positive oxidation state. In addition, since Br is bonded to N via a single bond, Br has an oxidation state of +1, indicating that the electron density of the electron contributed by Br to form the covalent bond lies towards N.)

(For more help, refer to Nitrogen Compounds tutorial)

(iii) 6 

(iv) The other product is CO_3^{2-}.

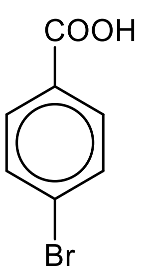
$R–N=C=O + 2OH^- \rightarrow R–NH_2 + CO_3^{2-}$ [4]

(e) <u>Step V</u>: Br_2 and $FeBr_3$ / Fe, room conditions

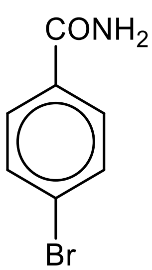
<u>Step VI</u>: acidified $KMnO_4$(aq), heat under reflux

<u>Step VII</u>: PCl_5 / PCl_3 / $SOCl_2$, room conditions

<u>Step VIII</u>: NH_3, room conditions



compound **C**



compound **D**

[7]

[Total: 20]

2 **(a)** **(i)** $pV = nRT$

$(1.00 \times 10^5)(386 \times 10^{-6}) = n\ (8.31)\ (298)$

$n = 1.56 \times 10^{-2}$ mol

Mass of gas = 1.00 – 0.438

= 0.562 g

Average M_r = 0.562 ÷ (1.56×10^{-2})

= 36.1

(ii) CO_2

(iii) M_r of CO_2 = 44.0

Let the M_r of the other gas be x.

mole faction of CO_2 = 193 / 386 = 0.5;

mole faction of other gas = 0.5

Average Mr = 0.5(44.0) + 0.5 x = 36.1

$x = 28.2 \approx 28$

Other gas is **<u>CO</u>**

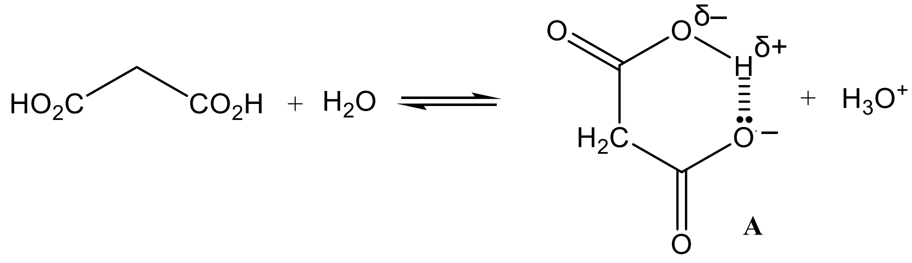
(iv) Since volume of CO_2 and CO produced is equal, $CO_2 \equiv CO$.

Hence, **E** is CaC_2O_4

$CaC_2O_4 \rightarrow CaO + CO_2 + CO$

(Verify that $CaO \equiv CO_2 \equiv CO$ by calculating the amount of CaO produced using the mass of white solid remaining.) [5]

(b) **(i)** Malonic acid is a stronger acid than ethanoic acid due to the **<u>stabilisation of the mono-anion</u>** by **<u>intramolecular hydrogen bonding</u>** with the unionised **<u>–CO_2H group</u>**.



A

(–O^- is attracted to δ+ H. Hence, the mono-anion is less likely to react with a H^+ ion to give back the undissociated acid.)

(ii) Removal of H^+ from an anion (**A**) that **<u>already carries a negative charge</u>** is **<u>electrostatically unfavourable</u>**.

OR

The stabilising hydrogen bonding would be destroyed by the ionisation of the second $–CO_2H$ group.

(iii) Let the acid be HA and the concentration of $H^+(aq)$ be x

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

$$10^{-2.85} = \frac{x^2}{0.10 - x}$$

Since x is small,

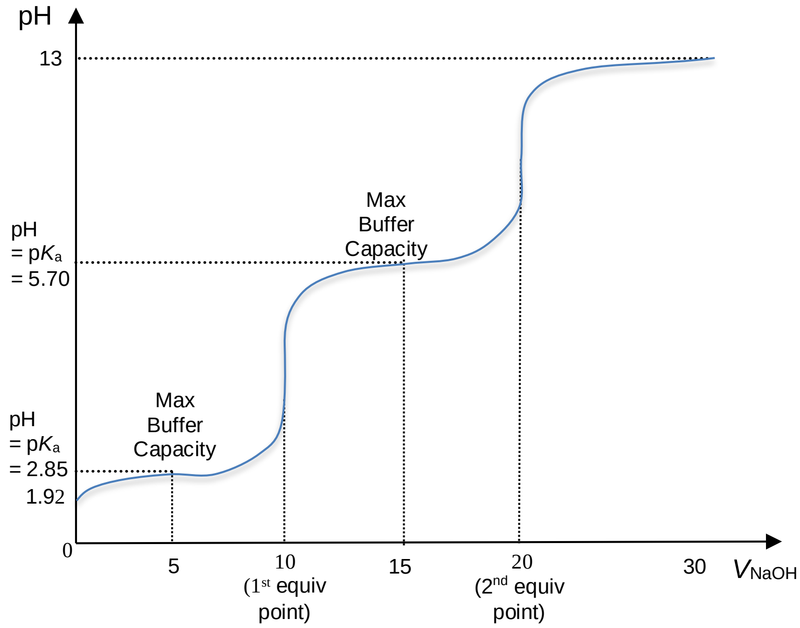
$$10^{-2.85} = \frac{x^2}{0.10}$$

$x = 1.19 \times 10^{-2}$ mol dm^{-3}

pH = –lg (1.19×10^{-2})

= 1.92

(iv)



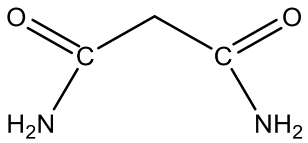
Key points:

- ✓ Initial pH
- ✓ Volume at the 2 equivalence points
- ✓ pH and volume at the 2 points of max buffering capacity
- ✓ curve flattens off between pH12 – 13 [7]

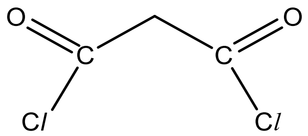
(c) (i) O=C=C=C=O

Linear

(ii) <u>(Electrophilic addition of C=C bonds) with NH_3</u>

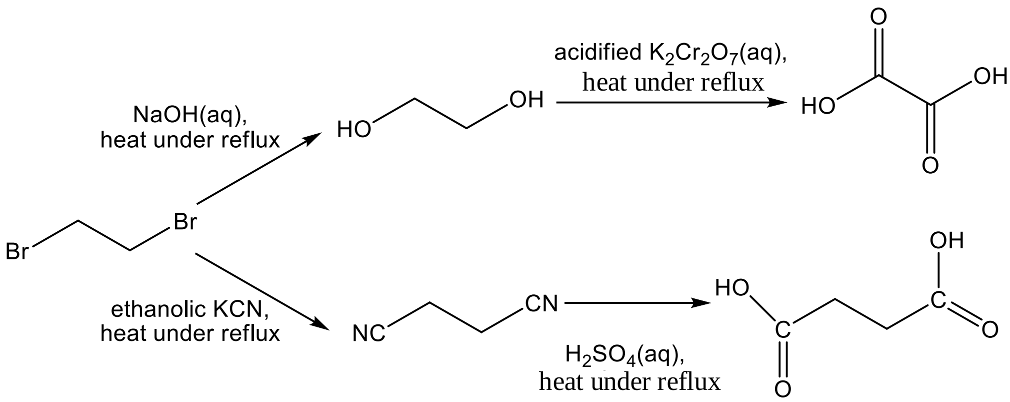


<u>(Electrophilic addition of C=C bonds) with HC*l*</u>



[4]

(d)



(To oxidise ethanediol to ethanedioic acid, $KMnO_4$(aq) CANNOT be used. This is because ethanedioic acid will be further oxidised to CO_2.)

(Draw the full structure formula showing all the C and H, if you are not confident / careful enough to draw skeletal formula.) [4]

[Total: 20]

3 **(a)** **(i)** $[Ca^{2+}] = 0.0080 \div 40.1$

$= 2.00 \times 10^{-4}$ mol dm^{-3}

$[Mg^{2+}] = 0.0049 \div 24.3$

$= 2.02 \times 10^{-4}$ mol dm^{-3}

$[Cl^{-}] = 0.0071 \div 35.5$

$= 2.00 \times 10^{-4}$ mol dm^{-3}

$[HCO_3^{-}] = 0.0366 \div (1.0 + 12.0 + 3 \times 16.0)$

$= 6.00 \times 10^{-4}$ mol dm^{-3}

The salts present are $MgCl_2$, $CaCl_2$, $Mg(HCO_3)_2$ and $Ca(HCO_3)_2$ in the relative amounts of 1:1:3:3.

(Other ratios are acceptable, as long as overall Ca^{2+}:Mg^{2+} = 1:1, and Cl^{-} : HCO_3^{-} = 1:3)

(ii) **$CaCO_3$**.

Partial evaporation of the mineral water caused **$[CO_2]$ to decrease**, as CO_2 is being removed during heating. This caused the **equilibrium position** of

$CO_2(g) + H_2O(l) + CO_3^{2-}(aq) \rightleftharpoons 2HCO_3^{-}(aq)$ to **shift to the left**. This results in an **increased $[CO_3^{2-}(aq)]$**.

Since **$CaCO_3$ is less soluble** than $MgCO_3$, the **K_{sp} value of $CaCO_3$ is lower** than that of $MgCO_3$.

The increased $[CO_3^{2-}(aq)]$ resulted in the **ionic product $[Ca^{2+}(aq)][CO_3^{2-}(aq)]$ to be higher than the K_{sp} of $CaCO_3$**, while the ionic product $[Mg^{2+}(aq)][CO_3^{2-}(aq)]$ is still lower than the K_{sp} of $MgCO_3$.

Thus, **$CaCO_3$** is the **first solid to be precipitated**.

(The solid is unlikely to be a metal chloride or metal hydrogencarbonate, since these salts tend to be soluble (bullet point 3). The solid is not a metal hydrogencarbonate, since it does not exist in the solid state (bullet point 4). Hence, it can be deduced that the solid is a metal carbonate, and you need to make use of the equilibrium reaction to suggest how CO_3^{2-} can be produced. Comparing $CaCO_3$ and $MgCO_3$, $CaCO_3$ is less soluble (bullet points 1 & 2).)

(Wrong: "$[H_2O]$ decreases, shifting equilibrium position to the left" – not accepted since H_2O is in a large excess in the *aqueous* solution, and its concentration will not change significantly due to evaporation.)

(iii) The rocks of hills and mountains probably contain **<u>$MgCO_3$, $CaCO_3$, $MgCl_2$ and $CaCl_2$</u>**.

<u>CO_2 dissolved in the rainwater</u> reacts with the **<u>CO_3^{2-} formed from the partial dissolution of the sparingly soluble $CaCO_3$ and $MgCO_3$</u>**, causing the **<u>equilibrium position</u>** of

<u>$CO_2(g) + H_2O(l) + CO_3^{2-}(aq) \rightleftharpoons 2HCO_3^-(aq)$</u> to **<u>shift to the right</u>**. This results in the **<u>formation of $Mg(HCO_3)_2(aq)$ and $Ca(HCO_3)_2(aq)$</u>** in mineral water.

As **<u>$MgCl_2$ and $CaCl_2$ are soluble</u>**, they will **<u>dissolve</u>** in the rainwater to form $MgCl_2$ (aq) and $CaCl_2$ (aq) in mineral water. [9]

(b) **(i)** $6Ca_5(PO_4)_3F + SiO_2 + H_2O \rightarrow SiF_4 + 2HF + 3CaO + 9Ca_3(PO_4)_2$

(ii) $Ca_3(PO_4)_2(s) \rightleftharpoons 3Ca^{2+}(aq) + 2PO_4^{3-}(aq)$

K_{sp} of $Ca_3(PO_4)_2 = [Ca^{2+}(aq)]^3[PO_4^{3-}(aq)]^2$ $mol^5\ dm^{-15}$.

(iii) Let $[Ca^{2+}(aq)]$ be x mol dm^{-3}.

Since $3Ca^{2+} \equiv 2PO_4^{3-}$,

$$[PO_4^{3-}(aq)] = \frac{2x}{3} \text{ mol dm}^{-3}$$

$$K_{sp} = x^3\left(\frac{2x}{3}\right)^2 = 1 \times 10^{-26}$$

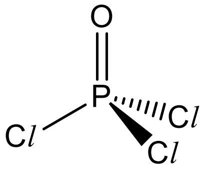
$x = 7.42 \times 10^{-6}$ mol dm^{-3} [5]

(c) **(i)** (Not in 9729 syllabus)

$P_4 + 6Cl_2 \rightarrow 4PCl_3$; **Heating** (white) phosphorus, P_4, with **<u>limited</u>** chlorine gas, Cl_2, to form PCl_3.

$P_4 + 10Cl_2 \rightarrow 4PCl_5$; **Heating** (white) phosphorus, P_4, with **<u>excess</u>** chlorine gas, Cl_2, to form PCl_5.

(ii) **G** is $POCl_3$.



Tetrahedral shape.

(iii) **H** is SO_2.

$$PCl_3 + SO_3 \rightarrow POCl_3 + SO_2$$

$$PCl_5 + H_2O \rightarrow POCl_3 + 2HCl$$ [6]

[Total: 20]

4 **(a)** (Not in 9729 syllabus)

The secondary structure of a protein is the **regular coils and folds** in **localised segments of the polypeptide chain**, as **stabilised by hydrogen bonding between *backbone* C=O and N–H groups of the polypeptide chain** (NOT C=O and N−H groups on the R groups).

The tertiary structure of a protein is the **overall three-dimensional shape of a polypeptide**, as **stabilised by side-chain / R group interactions** such as **hydrogen bonds**, **disulfide linkages**, **ionic interactions** and **van der Waals' forces**.

The quaternary structure of a protein is the spatial arrangement of **two or more polypeptide chain subunits**, **stabilised by side-chain / R group interactions** such as **hydrogen bonds**, **disulfide linkages**, **ionic interactions** and **van der Waals' forces** to form a large, complex protein molecule. [6]

(b) (Not in 9729 syllabus)

ΔG is **negative** since process is **spontaneous**.

ΔH is **negative** since **bonds are being formed releases energy** during the formation of the **secondary, tertiary and quaternary** structures.

ΔS is **negative** as the **disorder decreases** as the **polypeptide chains coil or come together**. (The polypeptide chains fold into a more ordered 3D structure) [3]

(c) **(i)** The electron configuration of Fe^{2+} is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$.

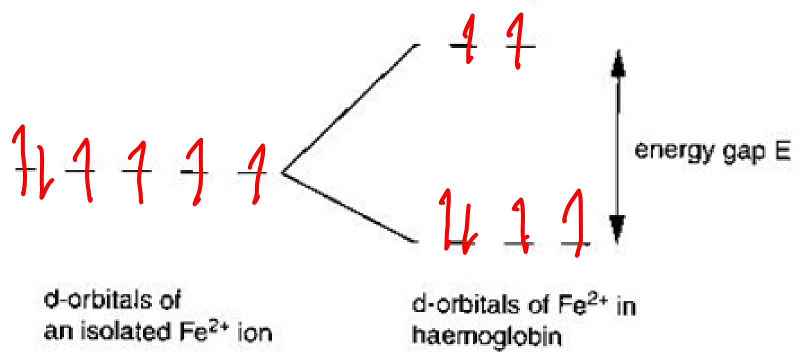
Fe^{2+} contains **partially filled 3d orbitals**. In the presence of H_2O or O_2 **ligands**, the **3d orbitals** are **split** into **2 sets** of non-degenerate orbitals with **different energies**. The difference in energies (**ΔE**) between these 2 sets of 3d orbitals is **small** and radiation from the **visible region** of the electromagnetic spectrum is absorbed when an **electron moves from a lower energy d-orbital to another unfilled/partially-filled d orbital of higher energy**. The colour observed (red) correspond to the **complement** of the absorbed colours (green). Hence, haemoglobin is coloured.

(Wrong:

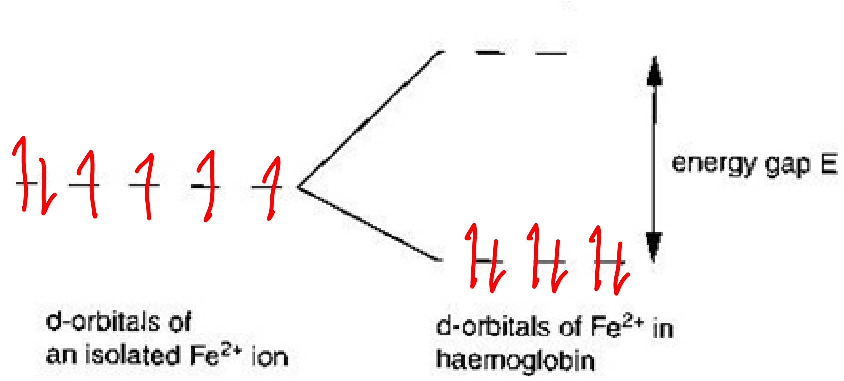
- "appear red due to the emission of red light" – red light is NOT emitted. The red colour is due to the *absorption* of green light.

- "promote the d-orbital from a lower energy level to a higher energy level" – it is the *electron* that is promoted to a d-orbital of higher energy level.)

(ii) **high spin state**



low spin state



(iii) Electrons usually occupy orbitals singly to **<u>minimise repulsion</u>** between **<u>two electrons occupying the same orbital</u>**.

(iv) The electrons would only pair up in the lower energy d-orbitals (low spin state) if the **<u>energy gap</u>**, ΔE, was **<u>larger</u>** than the **<u>energy due to electron repulsion</u>**.

Hence, low-spin **<u>oxyhaemoglobin</u>** would have the **<u>larger energy gap</u>**. [6]

(d) **(i)** Acid hydrolysis: $HCl(aq)$ or $H_2SO_4(aq)$, heat for several hours

OR

Alkaline hydrolysis: $NaOH(aq)$, heat for several hours.

(ii)

met	asp	gly							
	asp	gly	ser						
			ser	ala	gly				
					gly	glu	ser		
						glu	ser	lys	
							ser	lys	tyr

met-asp-gly-ser-ala-gly-glu-ser-lys-tyr [5]

[Total: 20]

5 **(a)** In order of increasing acidity: Ethanol < water < phenol

Ethanol is a weaker acid than water as the **electron-donating –C_2H_5 (ethyl)** group **intensify the negative charge on O**, making **$C_2H_5O^-$ (ethoxide ion) less stable**.

Phenol is a stronger acid than water as the **p orbital of oxygen overlaps** with the **π electron cloud of the benzene ring**. Hence, the **negative charge on O** is **delocalised into the benzene ring**, making **$C_6H_5O^-$ (phenoxide ion) more stable**. [4]

(b) Dilute HNO_3(aq) [1]

(c) 4-nitrophenol is **more acidic** than phenol as the **electron-withdrawing –NO_2 increases the delocalisation** of the **negative charge on O into the benzene ring**, thereby **stabilising the anion.** [1]

(d) **(i)** Step I 1) Sn, conc. HC*l*, heat

2) NaOH(aq)

(Sn is the reducing agent, not a catalyst!)

Step II NaOH(aq)

(Na(s) too reactive)

Step IV CH_3COCl

(ii) Step I Reduction

Step II Acid-base

Step III Nucleophilic substitution

Step IV Condensation

(iii) To produce an **anion**, **K** which is a **stronger nucleophile** than **J**. [8]

(e) Cl_2(aq) [1]

(f) **(i)** (Not in 9729 syllabus)

$3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClO_3 + 3H_2O$

(ii) $Fe^{2+} \rightarrow Fe^{3+} + e^-$

$ClO_x^- + 2\boldsymbol{x}H^+ + 2\boldsymbol{x}e^- \rightarrow Cl^- + \boldsymbol{x}H_2O$ (self-constructed)

$2\boldsymbol{x}Fe^{2+} \equiv 2\boldsymbol{x}e^- \equiv ClO_x^-$

Amount of Fe^{2+} = $11.3 \times 10^{-3} \times 0.500$

= 5.65×10^{-3} mol

Amount of ClO_x^- = $(5.65 \times 10^{-3}) / 2\boldsymbol{x}$

M_r of $KClO_x$ = 39.1 + 35.5 + 16$\boldsymbol{x}$ = 74.6 +16$\boldsymbol{x}$

Hence,

$(5.65 \times 10^{-3}) / 2\boldsymbol{x} = 0.150 / (74.6 + 16\boldsymbol{x})$

$\boldsymbol{x}$ = 2 (nearest whole number)

$4Fe^{2+}(aq) + ClO_2^-(aq) + 4H^+(aq) \rightarrow 4Fe^{3+}(aq) + Cl^-(aq) + 2H_2O(l)$

(Construct the respective half equations (ClO_2^- reduced to Cl^-, Fe^{2+} oxidised to Fe^{3+}), then combine to give the overall equation.)

[5]

[Total: 20]